

Seminarul 4

Sumatorul binar – completări

$$\begin{aligned}\bar{C}_i &= \overline{A_i B_i + (A_i + B_i) C_{i-1}} = \overline{A_i B_i} \cdot \overline{[(A_i + B_i) C_{i-1}]} = \overline{A_i B_i} \cdot (\overline{A_i + B_i} + \bar{C}_{i-1}) \\ &= \bar{G}_i (\bar{P}_i + \bar{C}_{i-1}) = \bar{G}_i \bar{P}_i + \bar{G}_i \bar{C}_{i-1} = \bar{P}_i + \bar{G}_i \bar{C}_{i-1},\end{aligned}$$

unde:

$$\bar{G}_i \bar{P}_i = \overline{A_i B_i} \cdot (\overline{A_i + B_i}) = (\bar{A}_i + \bar{B}_i) \bar{A}_i \bar{B}_i = \bar{A}_i \bar{B}_i + \bar{A}_i \bar{B}_i = \bar{A}_i \bar{B}_i = \bar{P}_i.$$